Interaction Obstructions in Einstein–Weyl Gravity: Cubic Protection, Second-Order Metric Failure, and Krein Visibility

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Abstract

We study whether the two Lorentz-covariant free completions of scalar-free Einstein–Weyl gravity extend to the interacting theory. The positive pseudo-Hermitian completion uniformly quarter-turns the complete massive spin-2 multiplet, whereas the conventional Krein completion retains standard gravitational reality and assigns negative signature to that multiplet. We first prove a cubic-protection theorem. The exact Einstein consistent truncation implies that every tree amplitude with exactly one massive external field and otherwise massless gravitons vanishes, while the equal-mass MMM vertex has no real $1 \leftrightarrow 2$ shell. Hence the first-order positive-metric obstruction is zero.

At second order the physical process $MM \to Mh$ is open. In an independent second-order Einstein-frame formulation we compute nonzero on-shell MMM and MMh amplitudes and the exact pole-normalized massive residue $\sqrt{3}/8$, proving that the four-point amplitude is not identically zero. A complete calculation in the original fourth-order theory — including the quartic contact term, all s, t, u exchanges, the physical adjoint, and gauge and equation-of-motion checks — gives at an interior rational physical point

$$\Pi_{\ker \mathrm{ad}_{h_0}}\left(T_{2,+}^{\dagger} - T_{2,+}\right) = -\frac{15762482064\, i}{5584765625}\, \sigma_x \neq 0.$$

The unique Lorentz-covariant positive free metric therefore admits no analytic second-order deformation compatible with the interaction.

We then classify conventional Poincaré-covariant, BRST-compatible, cluster-factorizing Krein gradings. Agreement with the free gravitational signature fixes their Fock lift to $J_F = (-1)^{N_M}$. The gravitational obstruction survives BRST cohomology and is not null:

$$\mathcal{O}^{\sharp} = \mathcal{O}, \qquad \mathrm{Tr}(\mathcal{O}^{\sharp}\mathcal{O}) = -8\mathcal{A}_K^2 \neq 0.$$

Unlike the perfect-square scalar theory, Einstein–Weyl gravity has no nontrivial uniform abelian charge grading capable of placing the obstruction in a one-sided null ideal. The result excludes both an analytic continuation of the canonical positive completion and the conventional massive-number grading as a null-sector probability mechanism. It does not exclude indefinite Krein evolution itself, nor nonlocal, non-factorizing, nonperturbative, or conformal-boundary completions.

*OpenAI model. Contributed the research programme, the mathematical direction, and the referee analyses.

†Anthropic model. Contributed the derivations, the machine verification (SymPy and exact rational arithmetic), and the manuscripts. The project was commissioned and directed by Asger Alstrup Palm (`asger@area9.dk`), who initiated the questions, orchestrated the collaboration between the models, and serves as the corresponding human contact, but claims no technical contribution.

Contents

1 Introduction

Scalar-free Einstein–Weyl gravity is the simplest four-dimensional quadratic-curvature theory whose flat-space physical spectrum consists of the ordinary massless graviton and one additional massive spin-2 multiplet. In the curvature convention used here its action is

$$S[g] = \int d^4x \sqrt{-g} \left(c_1 R + \alpha R_{\mu\nu} R^{\mu\nu} + \beta R^2 \right), \qquad \alpha = -3\beta, \tag{1}$$

with $M^2 = c_1/\alpha > 0$ in the split phase. Up to the four-dimensional Euler density, the quadratic term is Weyl-squared. The extra spin-2 branch has the opposite kinetic sign, the familiar higher-derivative ghost of quadratic gravity [1].

Paper IV of this series [13] did not choose between proposed interacting cures. It classified the *free* covariant completions. After symplectic reduction, all five massive polarizations form one irreducible massive spin-2 representation. Covariance therefore allows no helicity-by-helicity compromise. There are two real-form classes in the stated free, quasifree, mode-local setting:

1. a positive pseudo-Hermitian completion in which the entire massive multiplet has quarter-turned reality; and
2. a Krein completion with standard gravitational reality, positive massless signature, and uniform negative massive signature.

They determine the same free complex spectral functional but different adjoints and probability structures. The natural interacting question is therefore sharp:

Does either covariant free completion remain compatible with the full Einstein–Weyl interaction?

Paper V [14] supplied the deformation complex needed to ask the positive part of this question. If ρ_0 is the free Dyson map and v_n are the transported interaction vertices, an analytic correction $\rho_\zeta = \exp[-\frac{1}{2}\sum_n \zeta^n R_n]\rho_0$ must solve

$$
\begin{aligned}
[h_0, R_1] &= v_1^\dagger - v_1, \\
[h_0, R_2] &= (v_2^\dagger - v_2) + \frac{1}{2}[R_1, v_1 + v_1^\dagger].
\end{aligned}
\tag{2}
$$

On a degenerate free-energy shell the right-hand side must have zero projection onto $\ker \mathrm{ad}_{h_0}$. That shell projection is the metric obstruction. In the perfect-square scalar, first-order protection was followed by a second-order branch-changing obstruction; at the exact massless $O(1,1)$ boundary, however, the offending component could become one-sided in continuous charge and null in the generalized Born trace.

Gravity has the same perturbative-order pattern and a different mechanism:

$$
\boxed{\mathfrak{o}_+^{(1)} = 0, \qquad \mathfrak{o}_+^{(2)} \neq 0.}
\tag{3}
$$

The cubic protection follows from the exact Einstein subsector together with equal-mass kinematics. The second-order obstruction is carried by the physically open process

$$
M + M \longrightarrow M + h,
\tag{4}
$$

which changes massive particle number by one. The same process also separates gravity from the scalar boundary mechanism. Covariance and cluster factorization fix the conventional gravitational fundamental symmetry to $(-1)^{N_M}$, but $\mathbb{Z}_2$-odd does not imply null: the product of two odd operators is neutral and its quadratic trace is nonzero.

Results. The proof has four independent layers.

1. Section 3 proves a diagram-level phase theorem: a connected amplitude with E_M external massive legs changes by $(-i)^{E_M}$ under the uniform quarter-turn, including the compensating sign of every internal massive propagator.

2. Section 4 proves cubic protection. The Einstein consistent truncation kills Mhh, while MMM has no real physical $1 \leftrightarrow 2$ shell.

3. Sections 5–7 prove the second-order obstruction twice: first by an independent nonzero Einstein-frame factorization residue, then by a complete exact physical four-point calculation in the original fourth-order theory.

4. Sections 8–11 prove conventional Krein visibility: the covariant Fock grading is unique in the stated class, the exact block is not null, no scalar-like abelian charge exists, and the block is not BRST-exact.

Comparison with the scalar obstruction. The parallel and the difference are summarized by

Perfect-square scalar	Einstein–Weyl gravity
Källén zero and reality-factor protection	Einstein consistent truncation and equal-mass kinematics
$H + L \to L + L$	$M + M \to M + h$
$O(1,1)$ one-sided null ideal at the exact boundary	no nontrivial uniform abelian charge grading
obstruction can become probability-invisible	obstruction remains non-null for the conventional covariant Krein form

Strength of the claims. "Metric failure" means failure of the analytic order-by-order deformation of the canonical covariant positive free metric in the operator class of (2). It is not an absolute theorem against every positive metric, and no complex interacting spectrum is proved. "Krein visibility" has the precise finite-block meaning $\mathcal{O}^\sharp\mathcal{O} \neq 0$ and $\mathrm{Tr}(\mathcal{O}^\sharp\mathcal{O}) \neq 0$ for the unique conventional covariant, cluster-factorizing grading. It is a non-nullity statement, not a claim that an indefinite trace is a positive probability. In particular, the paper does not rule out an indefinite Krein dynamics; it rules out the naive parity/null-sector mechanism as a replacement for a positive interacting probability calculus.

Verification. Checks G13–G18 are exact SymPy or exact rational computations. They are mapped to the analytic statements in Appendix H. The exhaustive $5^3 \times 2$ polarization scan proposed as G16 is not used: the factorization residue, exact real-shell certificate, crossing, Ward, and gauge checks already close the proof. The scan remains optional regression hardening.

2 Free input, conventions, and the deformation class

2.1 Physical one-particle spectrum

We use signature $(+,-,-,-)$ in the main text. Around flat space the reduced physical one-particle space is

$$\mathcal{H}^{(1)}_{\mathrm{phys}} = \mathcal{H}_{h,+2} \oplus \mathcal{H}_{h,-2} \oplus \mathcal{H}_M, \qquad \dim \mathcal{H}_M = 5. \tag{5}$$

The massless summands carry helicities ± 2 on $p^2 = 0$; the massive summand carries the spin-2 representation on $p^2 = M^2$. Paper IV derives this by explicit diffeomorphism reduction rather than by a covariant propagator count. Its result is a healthy massless sector and a uniformly ghost-signed massive sector.

2.2 Two free real forms

In the positive class the original massive mass eigenfield is ‡-anti-Hermitian. We write the positive-frame observable as

$$\widehat{M}_{\mu\nu} = iM_{\mu\nu}, \qquad M_{\mu\nu} = -i\widehat{M}_{\mu\nu}. \tag{6}$$

The same quarter-turn acts on every one of the five polarizations. The massless graviton retains standard reality. In the conventional Krein class both fields retain standard gravitational reality, while the one-particle fundamental symmetry has signature

$$J^{(1)}_{\text{free}} = I_2 \oplus (-I_5), \qquad \text{signature } (+,+;-,-,-,-,-). \tag{7}$$

The uniformity is not optional. Since $\mathcal{H}_M$ is an irreducible massive spin-2 representation, Schur's lemma makes every covariant Hermitian form on its fiber a scalar multiple of the identity. Rotating only selected helicities would violate Lorentz covariance [13].

2.3 Finite-volume shell projection

As in Paper V, the cohomological equation is interpreted first at finite volume, where momenta are discrete and h_0 has point spectrum. If P_E projects onto free energy E, then

$$\Pi_{\ker \mathrm{ad}_{h_0}} S = \sum_E P_E S P_E, \tag{8}$$

and $[h_0, R] = S$ has a solution only if this projection vanishes. The rational physical point below can be placed on a momentum lattice after an overall choice of box scale. Continuum statements mean that the corresponding on-shell matrix element is nonzero on an open subset of the real scattering shell; a distributional rigged-Fock reformulation is not needed for the present existence theorem.

2.4 Amplitude convention

The perturbiner returns the multilinear coefficient of the action with the universal Feynman i stripped. We call it the reduced tree amplitude. At real momenta and real linear polarizations it is real. A global action rescaling and external polarization rescalings change the displayed rational certificate but cannot change its nonvanishing. All statements involving the precise coefficient below are therefore made in the conventions of Appendix E; the obstruction theorem itself is the invariant nonzero statement.

3 The external-massive-leg phase theorem

The quarter-turn phase must be transported through complete diagrams, not appended to an already computed external state by slogan. Internal massive lines carry the compensating kinetic sign.

Theorem 3.1 (External-massive-leg phase)**.** *Let $\mathcal{D}$ be a connected perturbative diagram written in mass-eigenfields (h, M), with E_M external massive legs and I_M internal massive lines. Under (6), its complete reduced amplitude obeys*

$$\mathcal{A}^{(E_M)}_{+}(\mathcal{D}) = (-i)^{E_M} \mathcal{A}^{(E_M)}_{K}(\mathcal{D}). \tag{9}$$

The statement includes contact terms, exchanges, and the sign of each massive inverse kernel.

Proof. Let n_v be the number of massive fields incident at vertex v. Field substitution contributes

$$\prod_v (-i)^{n_v}. \tag{10}$$

The massive quadratic form changes sign because $(-i)^2 = -1$; hence its inverse, the internal massive propagator, also contributes -1 per internal massive line. Every internal massive line is incident twice, so

$$\sum_v n_v = E_M + 2I_M. \tag{11}$$

The total factor is therefore

$$(-i)^{E_M+2I_M}(-1)^{I_M} = (-i)^{E_M}(-1)^{I_M}(-1)^{I_M} = (-i)^{E_M}. \tag{12}$$

Massless propagators are unchanged. Since the argument is diagramwise, it applies to their complete sum. Local invertible changes between regular mass-eigenfield frames can redistribute contact and exchange terms but leave the on-shell statement unchanged. □

Corollary 3.2 (Phase selection)**.** *For a real standard-frame tree amplitude, even E_M gives a real positive-frame coefficient and odd E_M gives a purely imaginary one. Only odd-E_M processes can contribute to the ordinary-adjoint anti-Hermitian shell source.*

For the four-point process (4), $E_M = 3$, and

$$\mathcal{A}_+ = i\mathcal{A}_K. \tag{13}$$

The sign is tied to the convention $M = -i\widehat{M}$; reversing the quarter-turn reverses the displayed sign but not the obstruction's nonvanishing.

4 Cubic protection

There are two odd-massive cubic structures: Mhh and MMM. The first is eliminated dynamically, the second kinematically.

4.1 Einstein consistent truncation

The scalar-free equations may be written schematically as

$$c_1 G_{\mu\nu} + \gamma_B B_{\mu\nu} = 0, \tag{14}$$

where $B_{\mu\nu}$ is the Bach tensor and γ_B depends on the normalization in (1). Every Ricci-flat Einstein metric has $G_{\mu\nu} = 0$ and is Bach-flat. More generally Einstein metrics are Bach-flat [3]. Thus the nonlinear Einstein solution space is an exact subsector of the Einstein–Weyl equations.

On the regular split branch, the higher-derivative theory admits a local second-order mass-eigenfield formulation in which the Einstein metric and the nonlinear massive spin-2 field are separate variables [2]. Denote them by (h, M) near flat space. The field map is algebraic and locally invertible on this branch, and $M = 0$ is precisely the Einstein subsector. These hypotheses are what make the following LSZ statement stronger than a basis-dependent observation.

Lemma 4.1 (One-massive-leg rule)**.** *For asymptotic mass eigenfields on the regular split branch,*

$$\mathcal{A}_{\rm tree}(M, h, \ldots, h) = 0. \tag{15}$$

In particular $\mathcal{A}_3(Mhh) = 0$ for every physical polarization.

Proof. Write the massive-field equation in the second-order frame as

$$\mathcal{K}_M M + J^{(2)}_M[h, h] + J^{(3)}_M[h, h, h] + \cdots = 0. \tag{16}$$

For every nonlinear vacuum Einstein solution $h_{\rm E}$, the pair $(h_{\rm E}, M = 0)$ solves the full theory. Hence the terms independent of M vanish on the Einstein graviton shell. Functional differentiation and LSZ reduction say that a physical amplitude with one external M and all remaining external legs in the Einstein subsector is zero. Terms generated by a local invertible change of fields are proportional to equations of motion or external inverse propagators and vanish under LSZ, the standard equivalence-theorem mechanism [4]. □

The exact perturbiner check evaluates the physical decay point with the massive particle at rest and two back-to-back massless gravitons. All $5 \times 4 = 20$ combinations of the five massive polarizations and four graviton-helicity choices vanish exactly. This is a check of the amplitude-level content of Lemma 4.1, not its only justification.

4.2 Equal-mass cubic kinematics

The MMM vertex itself is nonzero at complex on-shell kinematics (Section 5). It nevertheless has no real physical $1 \leftrightarrow 2$ shell. If $p_3 = p_1 + p_2$ with future-directed $p_i^2 = M^2$, then

$$p_3^2 = (p_1 + p_2)^2 \geq (2M)^2, \tag{17}$$

which cannot equal M^2. Equivalently, a particle of mass M cannot decay into two particles of the same mass.

Theorem 4.2 (Cubic protection)**.** *For scalar-free Einstein–Weyl gravity about flat space, the Lorentz-covariant positive real form has no first-order physical on-shell metric obstruction:*

$$\boxed{\mathfrak{o}^{(1)}_{+} = 0.} \tag{18}$$

Proof. By Corollary 3.2, only odd-massive cubic amplitudes can enter $\Pi_{\ker \mathrm{ad}_{h_0}}(v_1^\dagger - v_1)$. The Mhh amplitude vanishes by Lemma 4.1; the MMM amplitude has no real physical shell. The remaining cubic structures carry an even number of massive legs. Therefore the first-order anti-Hermitian source has zero physical shell projection. □

This protection is structurally different from the Källén-factor zero of Paper V. It is a consequence of an exact nonlinear consistent truncation plus massive-particle kinematics.

5 Independent Einstein-frame factorization rail

The next odd-massive process is four-point $MM \to Mh$. Before expanding the original fourth-order action to quartic order, we prove independently that this amplitude is not identically zero.

5.1 Second-order action and convention change

This section follows the exact Einstein-frame formulation of Magnano and Sokołowski [2]. Its verification artifact uses signature $(-,+,+,+)$, only in this section and Appendix B; thus a massive pole is $P^2 = -M^2$. With $M = 1$ and in the notation of that formulation, the nonlinear massive-field Lagrangian is

$$L_M = \frac{1}{2}K(Q) + \frac{m^2}{8A}\left(-\tau^2 + \psi_{\mu\nu}\psi^{\mu\nu} + 6A\tau - 12A^2\right), \qquad \psi = g + \Phi. \tag{19}$$

Here $Q^\alpha{}_{\mu\nu} = \Gamma^\alpha{}_{\mu\nu}(\psi) - \Gamma^\alpha{}_{\mu\nu}(g)$,

$$K(Q) = g^{\mu\nu}\left(Q^\alpha{}_{\mu\nu}Q^\beta{}_{\alpha\beta} - Q^\alpha{}_{\mu\beta}Q^\beta{}_{\nu\alpha}\right), \tag{20}$$

A is the determinant ratio and τ the corresponding trace defined in [2]. The form (19) separates Einstein gravity from a minimally coupled nonlinear massive spin-2 field.

For linearly traceless external fields, the determinant and inverse- determinant contributions to the apparent $\operatorname{tr}\Phi^3$ potential cancel exactly:

$$V(\Phi) = \frac{m^2}{8}\operatorname{tr}\Phi^2 + O(\Phi^4). \tag{21}$$

The nonzero MMM amplitude below comes entirely from $K(Q)$.

5.2 Exact three-point amplitudes

Take all momenta incoming. For the MMM vertex use

$$P = (1,0,0,0), \qquad p = \left(-\frac{1}{2}, 0, 0, \frac{i\sqrt{3}}{2}\right), \qquad q = \left(-\frac{1}{2}, 0, 0, -\frac{i\sqrt{3}}{2}\right). \tag{22}$$

Then $P+p+q = 0$ and $P^2 = p^2 = q^2 = -1$. With normalized TT polarizations (M_+, M_+, M_0), direct expansion of (19) gives

$$\boxed{\mathcal{A}_3(M_+, M_+, M_0) = -\frac{\sqrt{6}}{8}.} \tag{23}$$

For MMh, take

$$-P = (-1,0,0,0), \qquad r = (1,-1,-i,0), \qquad k = (0,1,i,0), \tag{24}$$

so $-P + r + k = 0$, $r^2 = -1$, and $k^2 = 0$. For the polarization choice in Appendix B,

$$\boxed{\mathcal{A}_3(M_+, M, h) = -\frac{\sqrt{2}}{8}.} \tag{25}$$

The graviton Ward identity vanishes for an arbitrary symbolic vector ξ_μ under $\varepsilon^h_{\mu\nu} \mapsto k_{(\mu}\xi_{\nu)}$, and exchange of the two massive legs leaves the coefficient unchanged.

5.3 Complete massive-pole residue

Glue (23) and (25) across P. A complete orthonormal rest-frame basis contains the polarizations $+, \times, 0, xz, yz$. The exact products are

λ	$\mathcal{A}_{MMM}$	$\mathcal{A}_{MMh}$	product
$+$	$-\sqrt{6}/8$	$-\sqrt{2}/8$	$\sqrt{3}/32$
$\times$	0	0	0
0	0	$\sqrt{6}/8$	0
xz	0	0	0
yz	0	$i\sqrt{2}/4$	0

Therefore the complete polarization numerator is

$$N = \sum_{\lambda=1}^{5} \mathcal{A}_3(M, M, M_P^{\lambda})\mathcal{A}_3(M_{-P}^{\lambda}, M, h) = \frac{\sqrt{3}}{32}. \tag{26}$$

The TT inverse kernel in the same action convention is

$$K_M(P) = \frac{P^2 + M^2}{4}. \tag{27}$$

Consequently

$$\boxed{\operatorname*{Res}_{P^2=-M^2} \mathcal{A}_4(M, M, M, h) = \frac{\sqrt{3}}{8} \neq 0.} \tag{28}$$

An overall rescaling of the action rescales the displayed residue but not its nonvanishing. A local quartic contact term is regular at this pole and cannot cancel it.

Proposition 5.1 (Factorization nonvanishing)**.** *The tree-level four-point amplitude tensor for $MM \to Mh$ is not identically zero on the relevant complexified on-shell component.*

Proof. If the rational amplitude vanished identically, every factorization residue would vanish. Equation (28) is the complete massive-polarization residue and is nonzero. □

6 The real physical shell

6.1 Open kinematics

In the center-of-mass frame, with signature $(+, -, -, -)$ restored,

$$|\mathbf{k}| = \frac{s - M^2}{2\sqrt{s}}, \qquad E_{M,\text{out}} = \frac{s + M^2}{2\sqrt{s}}, \qquad s \geq 4M^2. \tag{29}$$

Thus $MM \to Mh$ is physically open on a continuum shell, unlike the cubic MMM channel.

Lemma 6.1 (Complex nonvanishing and a real component)**.** *Fix analytic physical polarization sections on an irreducible component of the complexified four-point mass shell. The nonsingular real $MM \to Mh$ region contains a Euclidean-open center-of-mass chart and is Zariski dense in the complex component containing it. Hence a rational tree amplitude that vanishes on an open real subset vanishes identically on that component and has zero factorization residues.*

Proof. Away from exceptional polarization charts, the real physical region is parametrized by $s > 4M^2$, $-1 < \cos\theta < 1$, and an azimuth. Its real points contain an open set of maximal real dimension in the relevant algebraic component. If $\mathcal{A} = P/Q$ is regular there and vanishes on a real open set, real analyticity makes P vanish there. The identity theorem, equivalently Zariski density of that real locus, makes P vanish on the component. Then all residues vanish, contrary to (28). □

The next subsection supplies a stronger direct bridge: an exact nonzero point already on the real shell. It makes Lemma 6.1 logically redundant for the obstruction theorem, while retaining it as an independent explanation of why the complex factorization rail is physically relevant.

6.2 Interior rational certificate

Set $M = 1$. With p_1, p_2 incoming and p_3, k outgoing, take

$$\begin{aligned} p_1 &= \left(\frac{5}{4}, 0, 0, \frac{3}{4}\right), & p_2 &= \left(\frac{5}{4}, 0, 0, -\frac{3}{4}\right), \\ p_3 &= \left(\frac{29}{20}, -\frac{21}{25}, 0, -\frac{63}{100}\right), & k &= \left(\frac{21}{20}, \frac{21}{25}, 0, \frac{63}{100}\right). \end{aligned} \tag{30}$$

Then

$$p_1 + p_2 = p_3 + k, \qquad p_i^2 = 1, \qquad k^2 = 0, \tag{31}$$

and

$$s = \frac{25}{4} M^2, \qquad \cos\theta = \frac{3}{5}. \tag{32}$$

This point is interior and noncollinear. The planar kinematics has a y-reflection selection rule; the verification therefore uses explicit parity-even real linear polarizations rather than interpreting symmetry zeros as dynamical cancellations.

Computational Proposition 6.2 (Real-shell certificate, G15)**.** *For the real polarization tensors listed in Appendix E, the complete reduced tree amplitude at* (30) *is*

$$\boxed{\mathcal{A}_K(MM \to Mh) = \frac{7881241032}{5584765625} \neq 0.} \tag{33}$$

It is real, gauge independent, and nonsingular at this point. Therefore it is nonzero on a nonempty open neighborhood of the real physical shell.

7 Full fourth-order calculation and the metric obstruction

The calculation of Proposition 6.2 is performed directly in the original fourth-order variables of (1), independently of the Einstein-frame factorization rail.

7.1 Perturbiner and bordered propagator

For four plane waves,

$$g_{\mu\nu} = \eta_{\mu\nu} + \sum_{a=1}^{4} \lambda_a \varepsilon_{\mu\nu}^{(a)} e^{ip_a \cdot x}, \tag{34}$$

the wave algebra retains only multilinear subsets of the labels and implements $\partial_\mu \mapsto iP_\mu$ automatically. The coefficient of $\lambda_1\lambda_2\lambda_3\lambda_4$ in the action gives the exact quartic contact. The same engine produces the quadratic 10×10 operator $K(P)$ and cubic current J_A in the symmetric-tensor basis.

No symbolic closed propagator is assumed. In each s, t, u channel one solves the exact de Donder-bordered system

$$\begin{pmatrix} K(P) & C(P)^T \\ C(P) & 0 \end{pmatrix} \begin{pmatrix} X \\ \lambda \end{pmatrix} = \begin{pmatrix} -J^{(cd)} \\ 0 \end{pmatrix}, \qquad \mathcal{A}_{\text{exch}} = J^{(ab)T} X, \tag{35}$$

where

$$C_\mu(X) = P^\alpha X_{\alpha\mu} - \frac{1}{2} P_\mu X^\alpha{}_\alpha. \tag{36}$$

The exact field-equation residual and CX vanish in all three channels. Replacing C by the axial constraint $n^\mu X_{\mu\nu} = 0$ leaves the total amplitude unchanged.

7.2 Contact and exchange decomposition

At the rational point and polarization choice, the four contributions are

$$\begin{aligned} \mathcal{A}_{\text{contact}} &= \frac{8364585636057}{410644531250}, \\ \mathcal{A}_s &= -\frac{836906329}{45312500}, \\ \mathcal{A}_t &= -\frac{5142800561}{111695312500}, \\ \mathcal{A}_u &= -\frac{90855828716}{205322265625}. \end{aligned} \tag{37}$$

Their exact sum is (33). The contact term alone is not gauge invariant. Replacing the external graviton by $k_{(\mu}\xi_{\nu)}$ makes the *complete* contact-plus-exchange sum vanish for an arbitrary exact rational test vector; shifting a physical representative by the same gauge tensor leaves the amplitude unchanged. Initial-massive Bose exchange is exact. The two-point coefficient against every symmetric-tensor basis direction vanishes for each external on-shell polarization, excluding equation-of-motion contamination.

Reversing all external momenta and conjugating the real polarization sections reproduces the contact and each exchange separately. The quadratic kernels are recomputed and satisfy $K(-P) = K(P)$ exactly. Thus the reverse process entering the physical adjoint is evaluated, not inferred.

7.3 Quarter-turn of the complete operator

For the contact, three external massive fields give $(-i)^3 = i$. An exchange through an internal massless line has endpoint phase $(-i)^2(-i) = i$. An exchange through an internal massive line has endpoint phase $(-i)^3(-i)^2 = -i$, while the quarter-turned massive inverse kernel contributes -1, again giving i. Consequently every piece of (37) obeys Theorem 3.1, and

$$\langle Mh|T_{2,+}|MM\rangle = i\mathcal{A}_K, \qquad \langle MM|T_{2,+}|Mh\rangle = i\mathcal{A}_K. \tag{38}$$

Both states have exact free energy

$$E_{MM} = \frac{5}{4} + \frac{5}{4} = \frac{5}{2} = \frac{29}{20} + \frac{21}{20} = E_{Mh}. \tag{39}$$

On the ordered shell basis $(|MM\rangle, |Mh\rangle)$,

$$T_{2,+} = i\mathcal{A}_K\,\sigma_x, \tag{40}$$

and hence

$$\boxed{\Pi_{\ker \mathrm{ad}_{h_0}}\big(T_{2,+}^\dagger - T_{2,+}\big) = -2i\mathcal{A}_K\,\sigma_x = -\frac{15762482064\,i}{5584765625}\,\sigma_x \neq 0.} \tag{41}$$

7.4 First-order metric ambiguity

The solution R_1 of the first equation in (2) is defined up to a Hermitian G with $[G, h_0] = 0$. Such a change shifts the second-order source by $\frac{1}{2}[G, v_1 + v_1^\dagger]$. On the nonsingular real physical shell the complete connected cubic block is zero: Mhh vanishes by Lemma 4.1; MMM and MMh have no nonsoft real $1 \leftrightarrow 2$ shell; and real hhh three-point kinematics is collinear with zero physical helicity amplitude. Therefore

$$P_E(v_1 + v_1^\dagger)P_E = 0, \qquad P_E[G, v_1 + v_1^\dagger]P_E = 0. \tag{42}$$

The obstruction cannot be moved or canceled by choosing another analytic first-order metric representative.

Theorem 7.1 (Second-order positive-metric obstruction)**.** *The Lorentz-covariant positive free completion of scalar-free Einstein–Weyl gravity admits no analytic second-order metric deformation compatible with the full interaction on the physical $MM \to Mh$ shell. The obstruction is nonzero at* (30) *and on a nonempty open neighborhood of that point.*

Proof. Equation (41) lies in $\ker \mathrm{ad}_{h_0}$ and is nonzero. No R_2 can solve $[h_0, R_2]$ equal to this source. Equation (42) makes the class independent of first-order commutant freedom. The exact Ward, external-EOM, reverse-process, and internal-gauge checks show that the displayed matrix element is a reduced physical one. Since it is a nonsingular real-analytic amplitude and is nonzero at an interior point, it remains nonzero on an open neighborhood. □

8 Conventional covariant Krein grading

The positive obstruction does not by itself decide whether an indefinite completion can still be formulated. The narrower question addressed here is whether the conventional gravitational Krein grading can render the offending block null, as a continuous one-sided charge does at the perfect-square scalar boundary.

8.1 One-particle commutant

Let $J^{(1)}$ be a nondegenerate Hermitian involution on the physical one-particle cohomology (5). We require it to commute with the proper-orthochronous Poincaré representation and to agree with the free Krein real form of Paper IV.

The mass Casimir separates the massless and massive orbits, so no covariant endomorphism mixes $\mathcal{H}_h$ and $\mathcal{H}_M$. On the massive fiber, the exact spin-2 $SO(3)$ commutant is one-dimensional. There is one precision on the massless fiber: under the proper-orthochronous group, helicity $+2$ and helicity -2 are inequivalent complex representations, so covariance alone permits separate scalars. The complete commutant is therefore

$$J^{(1)} = \epsilon_+ I_{+2} \oplus \epsilon_- I_{-2} \oplus \epsilon_M I_5, \qquad \epsilon_\pm, \epsilon_M \in \{\pm 1\}. \tag{43}$$

Parity, or compatibility with the real gravitational field, exchanges the two massless helicities and imposes $\epsilon_+ = \epsilon_-$. Even without adding parity to the covariance group, agreement with the free signature (7) fixes

$$\epsilon_+ = \epsilon_- = +1, \qquad \epsilon_M = -1. \tag{44}$$

Proposition 8.1 (One-particle grading classification)**.** *Within the stated covariant class, the free real form uniquely fixes*

$$J^{(1)} = I_2 \oplus (-I_5). \tag{45}$$

No linear covariant grading can assign different signs within the massive spin-2 multiplet or mix the massive and massless shells.

8.2 Cluster-factorizing Fock lift

Let $j(n_h, n_M) \in \{\pm 1\}$ be the sign assigned to an asymptotic sector with n_h massless and n_M massive particles. Tensor multiplicativity and cluster decomposition require

$$j(n_h + n_h', n_M + n_M') = j(n_h, n_M) j(n_h', n_M'), \qquad j(0,0) = 1. \tag{46}$$

A homomorphism from the free commutative monoid $\mathbb{N}^2$ is fixed by its values on the two generators. Equations (44) and (46) therefore give

$$\boxed{J_F = (-1)^{N_M}.} \tag{47}$$

Assigning independent signs to unrelated multiparticle sectors would violate (46) and hence cluster factorization.

At the G17 shell,

$$J_F|MM\rangle = +|MM\rangle, \qquad J_F|Mh\rangle = -|Mh\rangle. \tag{48}$$

The transition crosses the two Krein signatures.

9 Krein visibility of the obstruction

For an operator on the positive-frame Hilbert representatives, define the Krein adjoint

$$X^\sharp = J_F X^\dagger J_F. \tag{49}$$

On the minimal ordered basis $(|MM\rangle, |Mh\rangle)$, $J_{\min} = \mathrm{diag}(1, -1)$.

Consider first the forward block

$$X = i\mathcal{A}_K \, |Mh\rangle\langle MM|. \tag{50}$$

It is J_F-odd:

$$J_F X J_F = -X. \tag{51}$$

Its Krein adjoint and quadratic product are

$$\begin{aligned} X^\sharp &= -i\mathcal{A}_K \, J_F |MM\rangle\langle Mh| J_F = i\mathcal{A}_K \, |MM\rangle\langle Mh|, \\ X^\sharp X &= -\mathcal{A}_K^2 |MM\rangle\langle MM|. \end{aligned} \tag{52}$$

Consequently

$$\boxed{\mathrm{Tr}(X^\sharp X) = -\mathcal{A}_K^2 = -\frac{621139602044804250 24}{311896070861816406 25} \neq 0.} \tag{53}$$

The sign is indefinite, but the operator is not null.

For the complete positive-metric obstruction

$$\mathcal{O} = -2i\mathcal{A}_K \sigma_x, \tag{54}$$

one obtains

$$\boxed{\mathcal{O}^\sharp = \mathcal{O}, \qquad \mathcal{O}^\sharp\mathcal{O} = -4\mathcal{A}_K^2 I_2, \qquad \mathrm{Tr}(\mathcal{O}^\sharp\mathcal{O}) = -8\mathcal{A}_K^2 \neq 0.} \tag{55}$$

In the exact rational convention,

$$-8\mathcal{A}_K^2 = -\frac{4969116816358434001 92}{3118960708618164062 5}. \tag{56}$$

Proposition 9.1 (Conventional Krein non-nullity)**.** *The physical $MM \to Mh$ block and the complete G17 obstruction are non-null for the unique conventional covariant, cluster-factorizing fundamental symmetry* (47)*. The obstruction is Krein-self-adjoint, indefinite, and visible to the finite-block quadratic trace.*

The algebraic distinction from a one-sided continuous grading is simple. For $\mathbb{Z}_2$,

$$1 + 1 = 0 \pmod 2. \tag{57}$$

The adjoint of an odd operator is odd and their product is even. Thus oddness supplies no charge-null lemma:

$$\boxed{J_F\text{-odd} \not\Longrightarrow \text{null.}} \tag{58}$$

By contrast, at the scalar $O(1,1)$ boundary a strictly one-sided nonzero continuous charge remains nonzero after the relevant product, and an invariant graded trace can kill it [14, 7].

Remark 9.2 (What is not ruled out). Equation (55) does not say that a Krein-self-adjoint interacting Hamiltonian cannot exist. Indeed, it displays Krein-self-adjointness of the minimal obstruction block. What fails is the stronger proposal that the conventional massive-number grading makes the branch-changing component null or supplies a positive generalized Born contribution. Indefinite pseudo-unitary evolution and a positive probability calculus are different questions.

10 No scalar-like charge-null mechanism

Could a continuous internal charge make the obstruction one-sided? Poincaré covariance first restricts any linear charge on the irreducible massive fiber to one scalar q_M. The free real graviton is neutral, $q_h = 0$. The verified nonzero interaction vertices impose

$$3q_M = 0 \quad (MMM), \qquad 2q_M + q_h = 2q_M = 0 \quad (MMh). \tag{59}$$

Proposition 10.1 (Uniform abelian charge no-go)**.** *There is no nontrivial uniform abelian charge assignment to the physical massive spin-*2 *irrep, with neutral graviton, that is respected by the Einstein–Weyl interaction.*

Proof. Equations (59) hold in any abelian charge group. Since $q_M = 3q_M - 2q_M$, they imply $q_M = 0$. The conclusion does not rely on torsion-freeness; $\gcd(2,3) = 1$ eliminates finite torsion charges as well. ☐

The MMM vertex has no real cubic decay shell but is a nonzero algebraic interaction vertex, so it already breaks massive-number parity at the action level. More decisively, the real G17 process itself changes N_M from 2 to 1 and proves

$$[(-1)^{N_M}, S] \neq 0 \tag{60}$$

on the physical scattering shell. Therefore the ingredients used to prove factorization simultaneously exclude an $O(1,1)$-like charge replacement.

11 BRST cohomology

The higher-derivative massive branch is a physical reduced mode and must not be confused with Faddeev–Popov ghosts. The external states in G17 are TT massive polarizations and TT massless graviton polarizations. The exact Ward identity, invariance under changes of gauge representative, axial/de Donder equality, and external inverse-propagator zeros show that the amplitude descends to reduced physical BRST cohomology.

Let Q denote the nilpotent BRST charge and let $|i\rangle, |f\rangle$ be closed physical representatives. If an operator were BRST-exact,

$$\mathcal{O} = \{Q, Y\}, \tag{61}$$

then, under the standard decoupling assumptions,

$$\langle f|\mathcal{O}|i\rangle = \langle f|QY|i\rangle + \langle f|YQ|i\rangle = 0. \tag{62}$$

The matrix element (41) is nonzero.

Proposition 11.1 (BRST survival)**.** *The G17 obstruction defines a nonzero operator on physical BRST cohomology and is not BRST-exact:*

$$[\mathcal{O}] \neq 0. \tag{63}$$

It cannot be placed in the standard BRST-null ideal.

Appendix G records the finite-complex algebra used in G18. In a splitting into physical cohomology and a contractible doublet, the physical-to-physical block of $\{Q, Y\}$ vanishes for a completely general Y, while the G17 block survives unchanged.

12 Master theorem

Theorem 12.1 (Interacting completion obstruction)**.** *Let scalar-free Einstein–Weyl gravity* (1) *be in its regular split phase about flat space. Use the covariant free completion classes of Paper IV and the analytic metric-deformation class of Paper V. Then:*

1. *the Lorentz-covariant positive pseudo-Hermitian completion is protected at first perturbative order,* $\mathfrak{o}_+^{(1)} = 0$*;*
2. *at second order the physical process* $MM \to Mh$ *generates the nonzero shell obstruction* (41)*;*
3. *hence the canonical covariant positive metric has no analytic second-order deformation compatible with the full interaction;*
4. *a conventional Poincaré-covariant, real-field-compatible, BRST-compatible, nondegenerate and cluster-factorizing Krein grading agreeing with the free real form has the unique Fock lift* $J_F = (-1)^{N_M}$*;*
5. *for this grading the obstruction is non-null,* $\mathrm{Tr}(\mathcal{O}^\sharp\mathcal{O}) = -8\mathcal{A}_K^2 \neq 0$*, and survives physical BRST cohomology; and*
6. *the nonzero* MMM *and* MMh *vertices prohibit a nontrivial uniform abelian charge grading analogous to the scalar* $O(1,1)$ *mechanism.*

Therefore neither an analytic deformation of the canonical positive metric nor the conventional massive-number null/parity mechanism supplies an interacting positive-probability completion of the full theory in the stated class.

Proof. Item 1 is Theorem 4.2. Items 2 and 3 are Theorem 7.1, independently supported by the factorization Proposition 5.1. Item 4 is Proposition 8.1 together with the monoid argument (46). Item 5 is Propositions 9.1 and 11.1. Item 6 is Proposition 10.1. □

13 Scope and nonclaims

The hypotheses in Theorem 12.1 are part of the result. The paper rules out:

- analytic deformation of the canonical Lorentz-covariant positive free metric;
- a conserved naive massive-number ghost parity;
- conventional covariant linear one-particle fundamental symmetries that agree with the free real form but differ on selected massive helicities;
- non-multiplicative independent Fock-sector signs, once cluster factorization is required;
- a uniform abelian continuous- or torsion-charge null mechanism compatible with the verified vertices; and
- standard BRST-exact nullity of the physical obstruction.

It does *not* prove:

- that no nonperturbative or differently pointed positive completion exists;
- that the full interacting Hamiltonian has complex energies;
- that every nonlocal or non-factorizing grading fails;
- that no singular $M \to 0$ or conformal-boundary completion can be defined;
- that a Maldacena-type boundary truncation of the solution algebra is inconsistent [8]; or
- that the result extends unchanged to quadratic gravities with a scalaron, matter, cosmological constant, or a different background.

Any surviving null-sector construction must therefore leave the conventional class. Possibilities include a grading singular at the conformal boundary, a nonlinear mixing of asymptotic particle sectors, a nonlocal observable algebra, a failure of ordinary cluster factorization, or boundary conditions that remove part of the fourth-order solution space. These are qualitatively different from a straightforward gravitational lift of the Bateman–Turok mechanism.

14 Discussion

14.1 The same order pattern, for different reasons

Paper V and the present gravity calculation share the sequence

$$\text{first-order protection} \longrightarrow \text{second-order scattering obstruction.} \tag{64}$$

The agreement in order should not obscure the difference in structure. The perfect-square scalar is protected by its special cubic kinematic factor and reality assignments. Gravity is protected because the Einstein solution space is an exact nonlinear subsector and the remaining odd cubic vertex is kinematically closed. In the scalar, an enhanced $O(1,1)$ charge appears exactly at the perfect-square boundary and can sort an obstruction into a one-sided null ideal. In gravity, the MMM and MMh interactions remove every uniform abelian charge before the four-point obstruction is considered.

14.2 Real forms become dynamical

At the free level a real form selects which complex solutions are called real and which adjoint defines observables. The choice can appear kinematic. The phase theorem makes its interacting content explicit: every physical process with odd external massive number crosses the two reality assignments and becomes anti-Hermitian in the positive frame. The decisive diagnostic is therefore

$$\boxed{\text{Which physical on-shell processes change the real-form sector?}} \tag{65}$$

For Einstein–Weyl gravity the first answer is $MM \to Mh$.

14.3 Why the factorization and physical rails are both useful

The Einstein-frame residue (28) is compact, independent, and immune to cancellation by local contact terms. It proves a genuine interaction before the large quartic calculation is attempted. The original-variable physical point then supplies what factorization alone does not: an exact real-shell value, the full gauge cancellation, the physical adjoint, and the deformation-complex matrix element. Agreement between the two rails is substantially stronger than either calculation in isolation.

14.4 What Krein visibility means

The negative traces in (53) and (55) are not advertised as negative probabilities. They show that the conventional quadratic Krein form sees the block: it is neither zero nor killed by a grading selection rule. A viable interacting probability construction would need extra structure beyond the free fundamental symmetry. The scalar boundary supplies such extra structure through a continuous one-sided charge; the verified Einstein–Weyl vertices do not.

The immediate research problem is consequently no longer another scalar doubling. It is to understand whether conformal-boundary conditions, changed observable algebras, or genuinely nonlocal gravitational structures can evade the hypotheses of Theorem 12.1 without destroying covariance and factorization.

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A Einstein consistent truncation and LSZ

This appendix spells out the hypotheses behind Lemma 4.1. For the scalar-free combination in (1), the field equation is the Einstein tensor plus a multiple of the Bach tensor. In four dimensions the Bach tensor may be written

$$B_{\mu\nu} = \left(\nabla^\rho\nabla^\sigma + \frac{1}{2}R^{\rho\sigma}\right) C_{\mu\rho\nu\sigma} \tag{66}$$

up to a convention-dependent overall sign. It vanishes on every Einstein metric [3]. In the flat-background problem the relevant exact subsector is Ricci-flat vacuum Einstein gravity.

On the regular branch of the Legendre transformation from Ricci-tensor gravity to its second-order form, introduce the Einstein metric and a symmetric massive field $\Phi_{\mu\nu}$. The transformation is algebraic in the auxiliary variables and locally invertible wherever its determinant does not vanish [2]. Linearization about the common flat vacuum diagonalizes the asymptotic equations into

$$\mathcal{K}_h h = 0, \qquad \mathcal{K}_M M = 0, \tag{67}$$

with $p^2 = 0$ and $p^2 = M^2$, respectively. Thus h and M used in the main text are LSZ mass eigenfields, not arbitrary components of the original metric perturbation.

Let $E_M[h, M] = 0$ be the exact massive-field equation in this frame. The consistent truncation states

$$E_M[h_E, 0] = 0 \tag{68}$$

for every nonlinear Einstein solution h_E. Expanding about flat space,

$$E_M[h, M] = \mathcal{K}_M M + \sum_{n\geq 2} J_M^{(n)}[h^n] + \sum_{r\geq 1, s\geq 1} J_{r,s}[M^r h^s]. \tag{69}$$

Equation (68) does not require each off-shell functional $J_M^{(n)}$ to vanish. It requires its value on a full perturbative Einstein solution to cancel, including nonlinear corrections to that solution. LSZ amputation removes precisely the external inverse- propagator and field-redefinition terms, leaving

$$\mathcal{A}(M, h_1, \ldots, h_n) = 0 \tag{70}$$

for physical on-shell Einstein gravitons. This is the standard consistent-truncation selection rule.

The perturbiner supplies an explicit cubic check. In signature $(+,-,-,-)$, take all momenta incoming,

$$p_M = (1,0,0,0), \qquad p_2 = \left(-\frac{1}{2}, 0, 0, -\frac{1}{2}\right), \qquad p_3 = \left(-\frac{1}{2}, 0, 0, \frac{1}{2}\right). \tag{71}$$

They obey $p_M^2 = 1$, $p_2^2 = p_3^2 = 0$ and sum to zero. The exact action coefficient vanishes for all five massive TT polarizations and the four products of the two graviton helicities. Replacing either graviton polarization by $p_{(\mu}\xi_{\nu)}$ also gives zero.

The selection rule is tree-level and concerns external mass eigenfields. It does not state that the original fourth-order metric action lacks terms apparently linear in a chosen massive component, nor does it exclude loop effects, anomalies, or a singular failure of the mass-eigenfield map at the conformal boundary.

B Einstein-frame cubic expansion

We record enough of the independent G14 calculation to make the exact numbers reproducible. This appendix uses signature $(-,+,+,+)$ and $M = 1$. It matches `symbolic/verify_gravity_factorization.py`.

For a symmetric matrix Φ, write

$$X = g^{-1}\Phi, \qquad t_n = \mathrm{tr}(X^n), \qquad A = \sqrt{\det(I+X)}. \tag{72}$$

Through cubic order,

$$\begin{aligned} A &= 1 + \frac{1}{2}t_1 + \frac{1}{8}t_1^2 - \frac{1}{4}t_2 + \frac{1}{48}t_1^3 - \frac{1}{8}t_1 t_2 + \frac{1}{6}t_3 + O(\Phi^4), \\ A^{-1} &= 1 - (A-1) + (A-1)^2 - (A-1)^3 + O(\Phi^4). \end{aligned} \tag{73}$$

In the potential of (19), $\tau = 4 + t_1$ and the relevant quadratic contraction is $4 + 2t_1 + t_2$. Substitution gives, when $t_1 = 0$,

$$A^{-1}\left(-\tau^2 + \psi_{\mu\nu}\psi^{\mu\nu} + 6A\tau - 12A^2\right) = t_2 + O(\Phi^4), \tag{74}$$

with zero t_3 coefficient. This is the potential cancellation (21).

At first order in one plane wave (p, E),

$$Q^{(1)\alpha}{}_{\mu\nu}(E,p) = \frac{i}{2}\eta^{\alpha\beta}\left(p_\mu E_{\beta\nu} + p_\nu E_{\beta\mu} - p_\beta E_{\mu\nu}\right). \tag{75}$$

The quadratic correction $Q^{(2)}$ comes from the inverse-ψ factor. The cubic MMM coefficient is the sum of all contractions $K(Q^{(1)}, Q^{(2)}) + K(Q^{(2)}, Q^{(1)})$, with the factor $1/2$ in (19). Exact substitution of (22) and normalized TT tensors gives (23).

For MMh, the script expands g^{-1}, ψ^{-1}, the connection of g, and Q through the orders required for the coefficient of three independent wave labels. At (24), take

$$v_r = (1,-1,0,0), \qquad e_h = (1,0,0,1), \qquad E_r = v_r^\flat \otimes v_r^\flat, \qquad H = e_h^\flat \otimes e_h^\flat. \tag{76}$$

Together with the internal + polarization at rest these tensors are transverse and traceless, and the coefficient is (25). For

$$H^{\text{gauge}}_{\mu\nu} = k_\mu \xi_\nu + k_\nu \xi_\mu \tag{77}$$

with four independent symbolic components of ξ, the coefficient vanishes identically.

Finally, the massive TT two-point coefficient for waves (p, E) and $(-p, E)$ with $E \cdot E = 1$ is

$$L^{(2)}_M = \frac{p^2 + 1}{4}. \tag{78}$$

This fixes the factor of four between the polarization numerator $\sqrt{3}/32$ and pole residue $\sqrt{3}/8$; it is not inferred from a unit-residue completeness convention.

C Polarization basis and factorization sum

At the rest momentum $P = (1, 0, 0, 0)$ in the Einstein-frame signature, let e_x, e_y, e_z be unit spatial vectors. A normalized massive basis is

$$\begin{aligned}
E_+ &= \frac{1}{\sqrt{2}}(e_x e_x - e_y e_y), & E_\times &= \frac{1}{\sqrt{2}}(e_x e_y + e_y e_x),\\
E_0 &= \frac{1}{\sqrt{6}}(e_x e_x + e_y e_y - 2 e_z e_z), & E_{xz} &= \frac{1}{\sqrt{2}}(e_x e_z + e_z e_x),\\
& & E_{yz} &= \frac{1}{\sqrt{2}}(e_y e_z + e_z e_y).
\end{aligned} \tag{79}$$

Each is transverse, traceless, and orthonormal under the induced tensor inner product. For a massive momentum $(E, 0, 0, p_z)$, replace e_z by the normalized longitudinal transverse vector; the same formulas give the transported helicity basis.

The factorization numerator is basis-independent. In a nonorthonormal basis T_a with Gram matrix $G_{ab} = T_a \cdot T_b$, it is

$$N = V_a (G^{-1})^{ab} U_b, \tag{80}$$

where $V_a = \mathcal{A}_3(M, M, T_a)$ and $U_b = \mathcal{A}_3(T_b, M, h)$. The orthonormal basis above reduces this to the sum displayed after (26). All five intermediate states are included; the result is not one selected nonzero channel.

The original fourth-order perturbiner supplies a separate complex factorization check at a rational-i point. It constructs an exact five-dimensional TT basis at the internal massive momentum, contracts the two cubic vectors with its Gram inverse, and obtains a nonzero reduced numerator. The Einstein-frame calculation then fixes a compact normalized representative and the explicit inverse-kernel slope.

D Physical-shell geometry and the rational point

For incoming equal masses in the center-of-mass frame,

$$p_1 = (E, 0, 0, p), \qquad p_2 = (E, 0, 0, -p), \qquad E = \frac{\sqrt{s}}{2}, \quad p = \frac{1}{2}\sqrt{s - 4M^2}. \tag{81}$$

Choose the outgoing graviton at scattering angle θ,

$$k = (\kappa, \kappa \sin\theta, 0, \kappa\cos\theta), \qquad \kappa = \frac{s - M^2}{2\sqrt{s}}, \tag{82}$$

and $p_3 = p_1 + p_2 - k$. Then $p_3^2 = M^2$ and $p_3^0 = (s + M^2)/(2\sqrt{s})$. The domain

$$s > 4M^2, \qquad -1 < \cos\theta < 1 \tag{83}$$

is a real open chart away from forward/backward and threshold degeneracies.

At $s = 25M^2/4$ one has $E = 5M/4$ and $\kappa = 21M/20$. Taking $\cos\theta = 3/5$ and $\sin\theta = 4/5$ gives exactly (30). All energy, momentum, and mass-shell identities are rational. Because all spatial y components vanish, reflection $y \mapsto -y$ is a symmetry. An odd number of y-odd polarization tensors forces an amplitude zero. The certificate uses the first nonzero all-even combination.

The exact threshold regression uses two incoming massive particles at rest and an outgoing graviton of energy $3M/4$. In the separate polarization convention of that test,

$$\mathcal{A}_{\text{threshold}} = -\frac{509784}{390625}. \tag{84}$$

Threshold is not used for the open-set claim; it guards the implementation against later changes.

E Full four-point engine and G17 expressions

E.1 Wave algebra

A wave-algebra element is a dictionary from subsets of external labels to sparse tensor components. Multiplication discards products in which a wave label repeats, so multilinear truncation is automatic. A derivative acting on subset S multiplies by

$$iP_{S\mu} = i \sum_{a\in S} p_{a\mu}. \tag{85}$$

From the metric, inverse metric, determinant, Christoffel symbols, and Ricci tensor, the engine extracts

$$\sqrt{-g}R, \qquad \sqrt{-g}R_{\mu\nu}R^{\mu\nu}, \qquad \sqrt{-g}R^2 \tag{86}$$

at the desired wave subset. Before any cubic or quartic claim it checks that the two-point coefficient vanishes on $p^2 = 0$ and $p^2 = M^2$, is nonzero off shell, and obeys cubic Ward identities.

E.2 Exact polarization tensors

For a massive momentum p in the (t, x, z) plane, choose two independent transverse in-plane vectors v_1, v_2. With

$$\Pi_{\mu\nu} = \eta_{\mu\nu} - \frac{p_\mu p_\nu}{M^2}, \tag{87}$$

the parity-even TT tensor used in the certificate is

$$E_{12,\mu\nu} = v_{1\mu}v_{2\nu} + v_{2\mu}v_{1\nu} - \frac{2}{3}(v_1 \cdot v_2)\Pi_{\mu\nu}. \tag{88}$$

The three massive external legs use the corresponding E_{12} at $p_1, p_2, -p_3$. For the graviton momentum direction proportional to $(5, 4, 0, 3)$, use transverse vectors

$$e_1 = (0, -3, 0, 4), \qquad e_2 = (0, 0, 1, 0), \tag{89}$$

and the real plus tensor

$$\varepsilon_+ = \frac{1}{25}e_1e_1 - e_2e_2. \tag{90}$$

These are the convention-fixed polarizations entering (33).

E.3 Gauge and adjoint checks

For each exchange, the exact bordered solution obeys

$$CX = 0, \qquad KX + C^T\lambda + J = 0. \tag{91}$$

The total amplitude with graviton gauge polarization is zero, while the contact contribution alone is

$$-\frac{596335802837691}{361367187500000} \neq 0. \tag{92}$$

This cancellation is a sensitive check of relative signs and combinatorics. De Donder and axial constraints give identical physical totals.

Under reversal $p_a \mapsto -p_a$, all contacts and cubic currents are recomputed. The action contains two- and four-derivative terms, and the independently recomputed quadratic matrices obey $K(-P) = K(P)$. Contact, s, t, and u pieces equal their forward values separately. Since the external linear tensors are real, this is the physical reverse matrix element used in $T^\dagger_{2,+}$.

E.4 Pole regression

An exact shell-preserving complex shift

$$p_1(z) = p_1 + z\eta, \qquad k(z) = k - z\eta, \qquad \eta = (3, 0, 4i, 5), \tag{93}$$

has $\eta^2 = \eta \cdot p_1 = \eta \cdot k = 0$. In the relevant channel

$$P(z)^2 = \frac{25}{4} + 15z \tag{94}$$

hits the massive pole at $z_* = -7/20$. The bordered kernel there has exactly the five TT zero modes. If T collects them and G_5 is their Gram matrix,

$$W = T^T K'(z_*)T = -\frac{15}{2}G_5 \tag{95}$$

entrywise, including 24 nontrivial identities. The residue

$$-j_1^T W^{-1} j_2 = -\frac{4551434384\, i}{328515625} \neq 0 \tag{96}$$

equals the independently normalized cubic contraction divided by the kinetic slope. A direct near-pole numerical probe of the actual bordered exchange agrees to relative error below 10^{-4}. This regression is not used to set the convention of the compact Einstein-frame residue; it validates the original-variable propagator machinery.

F Covariant grading and Fock-lift equations

G18a solves the commutant rather than assuming Schur's lemma numerically. At a standard massive momentum, represent the spin-2 fiber by symmetric traceless 3×3 tensors. The three $SO(3)$ generators act as

$$J_i(T) = L_i T + T L_i^T. \tag{97}$$

Solving $[X, J_i] = 0$ for a general 5×5 matrix gives $X = a_M I_5$. On the two complex massless helicity fibers, the little- group rotation generator is $\mathrm{diag}(2, -2)$, whose commutant is diagonal. Solving simultaneously with the mass Casimir on the full seven-dimensional fiber gives

$$X = \mathrm{diag}(a_+, a_-, a_M I_5). \tag{98}$$

Hermiticity and $X^2 = I$ restrict the three scalars to signs. The helicity-exchange real structure sets $a_+ = a_-$; the free signature fixes $(a_+, a_-, a_M) = (1, 1, -1)$.

For the Fock lift, cluster factorization is the monoid equation (46). Iterating it gives

$$j(n_h, n_M) = j(1, 0)^{n_h} j(0, 1)^{n_M} = (-1)^{n_M}. \tag{99}$$

The exact script verifies multiplicativity for all partitions in a finite table and the analytic monoid argument establishes it for all particle numbers.

G BRST block and non-nullity

The minimal non-null calculation uses

$$J_{\min} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad X = \begin{pmatrix} 0 & 0 \\ i\mathcal{A}_K & 0 \end{pmatrix}. \tag{100}$$

Then

$$X^\sharp = J_{\min} X^\dagger J_{\min} = \begin{pmatrix} 0 & i\mathcal{A}_K \\ 0 & 0 \end{pmatrix}, \qquad X^\sharp X = \begin{pmatrix} -\mathcal{A}_K^2 & 0 \\ 0 & 0 \end{pmatrix}. \tag{101}$$

For $\mathcal{O} = -2i\mathcal{A}_K \sigma_x$,

$$J_{\min} \mathcal{O}^\dagger J_{\min} = \mathcal{O}, \qquad \mathcal{O}^2 = -4\mathcal{A}_K^2 I. \tag{102}$$

To encode the cohomology argument, take a canonical finite splitting into two physical representatives and one contractible doublet $u \mapsto v \mapsto 0$. In the ordered basis (i, f, u, v),

$$Q = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}, \qquad Q^2 = 0. \tag{103}$$

For a completely general 4×4 matrix Y, direct multiplication shows

$$\big(QY + YQ\big)_{\mathrm{phys,phys}} = 0. \tag{104}$$

The upper-left physical block of the G17 obstruction is nonzero after embedding, so it cannot be of this exact form. This finite matrix is not a replacement for gravitational BRST quantization; it records the universal algebraic implication used after the Ward and reduced-helicity checks have established that the amplitude descends to cohomology.

H Machine-verification index

All scripts named below are in `symbolic/` and print `ALL PASS`. Checks use SymPy exact arithmetic except for the explicitly labelled near-pole regression.

Check	Claim	Artifact/type
G13a–b	two-point on-shell validation and cubic Ward identities	`gravity_perturbiner.py`, exact
G13	$Mhh = 0$ for 5×4 physical polarization choices	`verify_gravity_cubic.py`, exact
G14a–c	nonzero original-variable MMM, MMh, and reduced massive-pole factorization	`verify_gravity_cubic.py`, exact
G14d–i	potential cancellation, compact $-\sqrt{6}/8$ and $-\sqrt{2}/8$ amplitudes, symbolic Ward identity, five-state sum, inverse kernel	`verify_gravity_factorization.py`, exact
G15a–f	full physical contact+s, t, u, Ward, Bose, gauge, and threshold checks	`verify_gravity_g15.py`, exact
G15g	shifted-pole kernel, Schur slope, residue; near-pole probe	same script, exact plus numerical regression
G16	crossing needed for the adjoint is included in G17; exhaustive 250-polarization scan is deferred	not a theorem dependency
G17a–d	exposed contact/exchanges, reverse process, EOM/Ward/gauge, termwise quarter-turn	`verify_gravity_obstruction.py`, exact
G17e–f	exact shell obstruction and first-order metric-ambiguity independence	same script, exact
G18a–b	one-particle commutant and cluster-factorizing Fock lift	`verify_gravity_krein.py`, exact
G18c–e	non-null quadratic traces, charge no-go, and BRST block	same script, exact

The analytic inputs not reducible to finite symbolic identities are identified in the main text: the induced-representation interpretation of the free commutant (Paper IV), the consistent-truncation/LSZ argument, real-analytic continuation on the physical shell, and the standard BRST decoupling statement. The symbolic checks test every convention-sensitive coefficient consumed by those arguments.

Reproduction commands. From the repository root for this project:

```
python3 symbolic/verify_gravity_completion.py
python3 symbolic/verify_gravity_cubic.py
python3 symbolic/verify_gravity_factorization.py
python3 symbolic/verify_gravity_g15.py
python3 symbolic/verify_gravity_obstruction.py
python3 symbolic/verify_gravity_krein.py
```

The full G15 script contains a slower symbolic shifted-pole step; G17 recomputes the exact physical certificate and the checks needed by the master theorem in a compact import-safe four-point assembly.